

# DBMS Project Synopsis

## 1. Title Page

**Project Title:** Real-time Multi-Modal Electronic Toll Collection and Fraud Detection System (FASTag, NHAI Pass, and Cash)

**Course Name & Code:** UCS310 – Database Management Systems

**Degree & Year:** B.Tech (2nd Year)

**Department / Institute Name:** Computer Science Engineering Department, Thapar Institute of Engineering and Technology.

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**Academic Year:** 2025-26

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## 2. Introduction

Modern toll collection systems in India have evolved from simple manual collection to a complex, multi-layered ecosystem. This project designs a robust **Database Management System (DBMS)** capable of handling multiple payment modes: **FASTag (RFID)**, **NHAI Monthly Passes (200-trip limit)**, and **Manual Cash Payments**.

The system's core is built to prioritize automated digital payments while maintaining a fallback mechanism for manual transactions. By leveraging SQL and PL/SQL, the database ensures that every vehicle is processed according to its available credentials—checking for a valid FASTag first, then a valid NHAI pass, and finally defaulting to cash. This hierarchical processing requires sophisticated data modeling to ensure speed, accuracy, and fraud prevention across all payment modes.

### 3. Problem Statement

Managing multiple payment streams introduces significant database challenges:

- **Hierarchical Logic:** The system must instantly decide which payment source to charge (FASTag vs. Pass vs. Cash) without causing traffic delays.
- **Trip Tracking:** NHAI passes must accurately track and decrement the 200-trip quota in real-time, ensuring no usage beyond the limit or expiry.
- **Fraud across Modes:** Detecting fraudulent activities such as "Pass Sharing" (using one pass for multiple vehicles) or "Double Dipping" (charging both FASTag and a pass).
- **Cash Integrity:** Ensuring manual cash transactions are correctly logged and matched with vehicle detection data to prevent revenue leakage.

### 4. Objectives of the Project

- **Multi-Modal Data Modeling:** To design a schema that integrates **RFID Tags, FASTag Wallets, NHAI Monthly Passes, and Cash Transaction Logs.**
- **Hierarchical Payment Logic:** To implement PL/SQL procedures that automatically check payment eligibility in the order: FASTag -> NHAI Pass -> Cash.
- **Real-time Quota Management:** To automate the decrementing of trip counts for NHAI passes and balance updates for FASTag wallets.
- **Integrated Fraud Detection:** To identify anomalies like cloned tags, shared passes, and discrepancies in manual cash entries.
- **Normalization:** To organize the data up to 3NF to ensure efficient storage of diverse payment records.

### 5. Scope of the Project

The project covers the following modules:

- **FASTag Module:** RFID tag mapping and wallet balance management.
- **NHAI Pass Module:** Management of 200-trip monthly passes, including trip tracking and expiry validation.
- **Manual Toll Module:** Logging cash payments for vehicles without digital credentials.
- **Hierarchical Processing Engine:** The core logic that selects the appropriate payment mode.
- **Fraud Detection Engine:** Cross-referencing all payment modes to detect inconsistencies.

### 6. Proposed System Description

The proposed system acts as a centralized decision engine at the toll plaza. When a vehicle is detected:

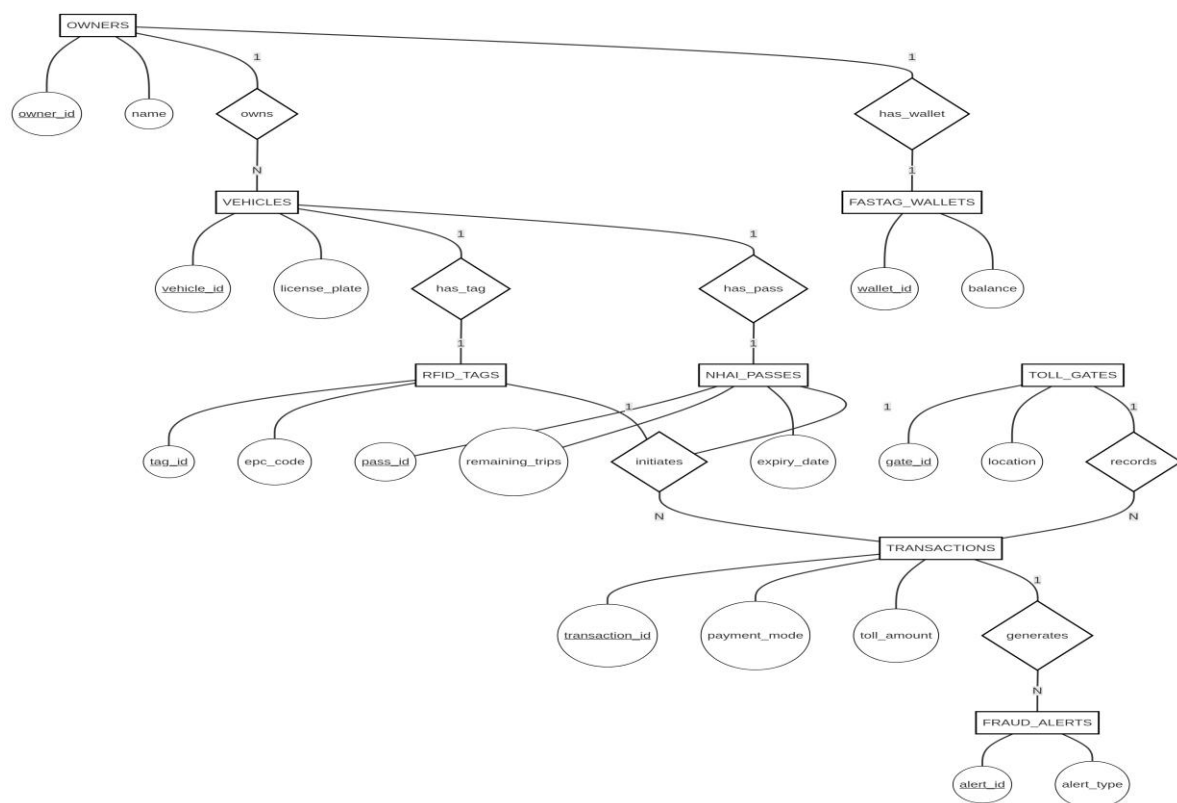
- 1 **RFID Scan:** The system first attempts to read the FASTag. If valid and funded, the toll is deducted.
- 2 **Pass Validation:** If no FASTag is found or it has insufficient funds, the system checks if the vehicle has an active **NHAI Monthly Pass** with remaining trips.
- 3 **Cash Default:** If neither digital mode is available, the system prompts for a **Cash Payment**, logs the transaction, and opens the barrier.

All these steps are governed by **PL/SQL Triggers** that validate data integrity and **Stored Procedures** that handle the multi-step payment logic atomically.

## 7. Database Design

### 7.1 Entity–Relationship (ER) Diagram (Chen's Notation)

The diagram now includes **NHAI\_PASSES** and the multi-modal attributes in the **TRANSACTIONS** entity:



## 7.2 Relational Schema

| Table Name     | Attributes (Primary Key, Foreign Key)                                                                                             |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------|
| OWNERS         | <b>&lt;u&gt;owner_id&lt;/u&gt;</b> , name, address, contact_info                                                                  |
| VEHICLES       | <b>&lt;u&gt;vehicle_id&lt;/u&gt;</b> , owner_id (FK), license_plate (Unique), vehicle_type                                        |
| FASTAG_WALLETS | <b>&lt;u&gt;wallet_id&lt;/u&gt;</b> , owner_id (FK), balance, status                                                              |
| RFID_TAGS      | <b>&lt;u&gt;tag_id&lt;/u&gt;</b> , vehicle_id (FK), epc_code (Unique), status                                                     |
| NHAI_PASSES    | <b>&lt;u&gt;pass_id&lt;/u&gt;</b> , vehicle_id (FK), remaining_trips, expiry_date, status                                         |
| TOLL_GATES     | <b>&lt;u&gt;gate_id&lt;/u&gt;</b> , location, lane_count                                                                          |
| TRANSACTIONS   | <b>&lt;u&gt;transaction_id&lt;/u&gt;</b> , vehicle_id (FK), gate_id (FK), payment_mode (FASTag/Pass/Cash), toll_amount, timestamp |
| FRAUD_ALERTS   | <b>&lt;u&gt;alert_id&lt;/u&gt;</b> , transaction_id (FK), alert_type, status                                                      |

## 8. Normalization

- **1NF:** All attributes like payment\_mode and remaining\_trips are atomic.
- **2NF:** remaining\_trips is fully dependent on pass\_id.
- **3NF:** Vehicle and Owner details are decoupled from specific payment credentials (Passes/Tags) to allow one owner to manage multiple vehicles and payment modes efficiently.

## 9. Database Implementation

### 9.1 SQL Implementation

- **DDL:** CREATE TABLE NHAI\_PASSES with a CHECK constraint on remaining\_trips  $\geq 0$ .
- **DML:** UPDATE NHAI\_PASSES SET remaining\_trips = remaining\_trips - 1 WHERE pass\_id = :pid.
- **Complex Queries:** SELECT payment\_mode, COUNT(\*) FROM TRANSACTIONS GROUP BY payment\_mode to analyze revenue distribution.

## 9.2 PL/SQL Components

- **Procedure: Process Multi Modal Toll:** Implements the logic:

```
IF FASTag_Valid THEN Deduct_Wallet;
```

```
ELSIF Pass_Valid THEN Decrement_Pass_Trips;  
ELSE Process_Cash;
```

```
END IF;
```

- **Trigger: Prevent Pass Misuse:** Ensures a pass linked to one vehicle\_id isn't being used by another vehicle (detected via license plate recognition).
- **Trigger: Low Trip Alert:** Automatically flags passes with fewer than 5 trips remaining.

## 10. Transaction Management

The system uses **SAVEPOINTS** within the hierarchical payment procedure. If a FASTag deduction fails, the transaction rolls back to the savepoint before trying the NHAI Pass, ensuring the database state remains consistent throughout the decision process.

## 11. Tools & Technologies

- **DBMS:** Oracle 19c / MySQL 8.0
- **Language:** SQL, PL/SQL
- **Design:** Extended Chen's Notation for Multi-Modal Systems

## 12. Expected Outcomes

- A unified database capable of managing FASTag, NHAI Passes, and Cash.
- Automated trip-tracking for 200-trip monthly passes.
- Reduced manual errors through hierarchical payment automation.
- Enhanced fraud detection across all transaction types.

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## References

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IJSART. (2025). *Automated Toll Tax Collection System Using Cloud Database*. Retrieved from <https://ijsart.com/public/storage/paper/pdf/IJSARTV11I4102966.pdf> [4] Soliz, J. (n.d.). *AI-Driven Fraud Detection in Electronic Toll Systems*. Retrieved from <https://josephsoliz.com/ai-driven-fraud-detection-in-electronic-toll-systems/>